

Palavalasa Neha

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EDUCATION

BMS Institute of Technology and Management

Dec 2021 – May 2025

Bachelor of Engineering in Information Science and Engineering, CGPA: 7.89

EXPERIENCE

Associate Analyst – AI Data

Apr 2026 – Jun 2026

GlobalLogic

- Built and maintained Python-based data annotation workflows for robotics and foundation-model training datasets, using temporal segmentation and semantic labeling of human-action video.
- Performed structured spatial and kinematic annotation of object manipulation, reaching, repositioning, and human-action trajectories to support machine-learning model development and evaluation.
- Executed dataset quality-control and validation checks for tracking failures, field-of-view issues, and camera artifacts; documented errors and improved training-data reliability for downstream AI pipelines.

System Engineer Trainee

Dec 2025 – Mar 2026

Infosys

- Developed Java-based backend services and REST APIs using layered architecture and microservices principles.
- Integrated SQL databases and wrote JUnit tests to validate business logic, API behavior, and application reliability.
- Participated in Agile SDLC activities including sprint planning, retrospectives, code reviews, debugging, and collaborative development.

INTERNSHIP

Software Engineering Intern

Mar 2025 – Jun 2025

RocketGrowth Technologies

- Developed a real-time AI-powered CCTV monitoring system using Python, YOLOv8, and OpenCV for automated worker and inventory detection in warehouse environments.
- Integrated LLM-powered decision support into a disaster-management dashboard by designing prompt pipelines and connecting model outputs with dynamic application components.
- Implemented QR/barcode recognition, event logging, PostgreSQL persistence, and zone-level tracking; collaborated using Agile/Scrum, JIRA, and Confluence.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL

AI / LLM: LangChain, Groq API, Llama 3.3, OpenAI API, Prompt Engineering, LLM Integration

ML & Data: TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, YOLOv8, Matplotlib

Backend & APIs: FastAPI, Flask, Spring Boot, Node.js, REST APIs, SQLAlchemy, JWT/OAuth

Databases: PostgreSQL, MySQL, MongoDB, Redis, ChromaDB

Cloud / DevOps: Docker, Kubernetes, Jenkins, CI/CD, AWS, Git

Web: ReactJS, GraphQL, HTML5, CSS3

Core CS: Data Structures, OOP, Computer Networks, Microservices, Agile, SDLC

PROJECTS

AI Ops Assistant – Multi-Agent System | Python, FastAPI, Groq (Llama 3.3), LangChain, REST APIs

- Architected Planner, Executor, and Verifier agents for LLM-driven task decomposition, tool execution, and response validation through a FastAPI REST endpoint.
- Integrated Groq-hosted Llama 3.3 and designed prompt pipelines for multi-step planning and verification.
- Built modular integrations with Open-Meteo and GitHub Search APIs for extensible tool-using workflows.

AI-Powered CCTV Monitoring System | Python, OpenCV, YOLOv8, PyZbar, PostgreSQL, MongoDB

- Developed real-time worker and inventory tracking with YOLOv8 and QR/barcode recognition; integrated LLM decision support for a disaster-management dashboard.
- Designed modular detection, logging, and analytics layers with PostgreSQL-based zone-level traceability.

Cloud-Based Modular E-Commerce Platform | ReactJS, Redux, GraphQL, Node.js, Module Federation, AWS

- Built a scalable micro-frontend e-commerce platform and integrated GraphQL; reduced API response times by 75% and improved front-end performance by 89%.

CERTIFICATIONS

- Oracle Certified Professional – Oracle Cloud Infrastructure: Generative AI (2024) – Oracle
- Spark and Python for Big Data with PySpark – Udemy, Jul 2024
- Java Certified Foundations Associate: Development Basics – Infosys Springboard
- Python Data Structures and Algorithms – Infosys Springboard